

IndyGo Ridership PowerBI Dashboard

Enhancing Public Transit through Data-Driven Insights

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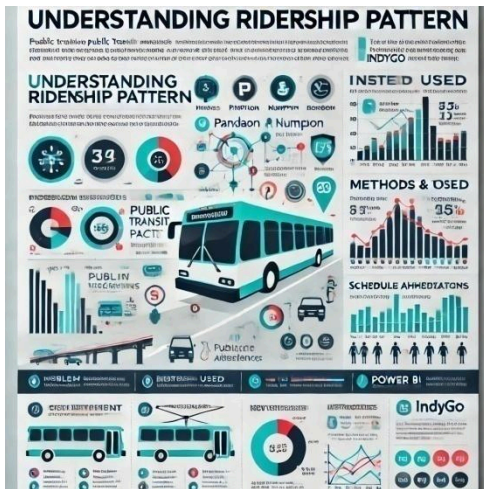


Fig. 1(a): A summary showcasing the problem, data types, methods, tools, and key visualizations.



Fig. 1(b): A continuous cycle from stakeholder needs to insights and feedback for refinement.

1. INTRODUCTION AND PRIOR WORK

1.1 Problem Space: Public transit is essential to urban mobility, yet transit agencies often struggle to analyze ridership data, optimize routes, and maintain transparency. IndyGo, Indianapolis' public transit provider, collects extensive data through GPS Trackers, APCs, and scheduling systems but transforming this data into actionable insights remains a challenge. To address this, our team is developing a Power BI Ridership Dashboard that visualizes trends across timeframes, routes, and stops. The dashboard will support transit planners, city officials, policymakers, and the public in understanding usage patterns, identifying high-demand areas, and improving service planning. This aligns with IndyGo's mission to enhance engagement, improve decision-making, and foster innovation in transit optimization.

1.2 Prior Work: Public transit agencies have developed various dashboards to visualize ridership data and support planning decisions. Notable examples include systems by the Chicago Transit Authority (CTA) [1], CapMetro [2], and MTA [3]. These platforms provide insights into trends, service reliability, and fleet performance through time-series charts, bar graphs, and summary metrics. Among them, CTA's Performance Dashboard [1] stands out for its comprehensive tracking of ridership and operational metrics. However, existing tools often lack granular stop-level analysis and real-time predictive capabilities, limiting their usefulness for dynamic service adjustments. This project builds on such efforts by incorporating more detailed, interactive, and localized insights to better inform daily transit operations and long-term planning. Similarly, CapMetro's Performance Dashboard [2] emphasizes monthly ridership trends, route-level analysis, and safety statistics. While useful for tracking long-term patterns, its static reporting limits dynamic data exploration. MTA Metrics Platform [3] offers a real-time view of daily ridership across subway and bus services, allowing users to explore current and historical data. Although powerful, its interface can be overwhelming for general users, and it lacks detailed route-specific comparisons needed for optimizing individual lines. Beyond agency dashboards, academic studies such as those on Seattle's South Lake Union Streetcar [4] and Salt Lake City's TRAX Light Rail [5] highlight the value of long-term trend analysis using historical data and statistical modeling. However, these studies are less applicable to real-time decision-making and operational planning. Together, these examples demonstrate strengths and gaps in existing transit visualizations, informing our project's focus on interactive, stop and route-level insights that support both historical analysis and daily operational decisions.

1.3 Advancing Beyond Existing Work: While these existing solutions provide valuable insights, they fall short in offering a fully

integrated and interactive analysis experience. The IndyGo Ridership Dashboard builds upon prior work by incorporating real-time analytics, route-based evaluations, and historical comparisons, enabling deeper insights into service demand. Unlike static reporting tools, our dashboard emphasizes interactive data visualizations, dynamic filtering to enhance decision-making. Additionally, it integrates multi-dimensional analysis, allowing stakeholders to compare ridership trends across various timeframes, route categories, and geographic regions. By addressing the limitations of prior dashboards, this project introduces an enhanced approach to transit analytics, ensuring a more comprehensive, user-friendly, and insightful experience for transit planners, policymakers, and daily commuters.

1.4 Research Questions: Our project will focus on analyzing IndyGo's ridership patterns using historical transit data to derive actionable insights for service optimization, operational improvements, and public engagement. The following key research questions will be addressed:

1.4.1 Ridership Trends over Time: What is the total ridership by day, week, and month? How does ridership change across different timeframes (e.g., monthly, or over the last five weeks)? What are the changes in ridership trends year-over-year? These questions will help identify seasonal trends, peak and low ridership periods, and long-term ridership shifts.

1.4.2 Route-Based Ridership Analysis: How does ridership vary across different route categories (e.g., high-frequency routes, medium-frequency routes, low-frequency routes)? What are the ridership trends for specific routes over the past 12 months? This analysis will allow transit planners to determine which routes require service adjustments and optimize frequency for rider demand.

1.4.3 Stop-Level Analysis: Which bus stops experience the highest and lowest ridership? This will help in identifying high-demand transit hubs and underutilized stops to enhance efficiency in service planning.

1.4.4 Comparative Analysis and Optimization: How does ridership compare across different service periods (e.g., weekdays vs. weekends, peak vs. off-peak hours)? This will support decision-making for better schedule planning and improved service reliability.

1.5 Stakeholders and Their Insight Needs: The project identifies three key stakeholder groups: IndyGo Transit Planners, Public Transit Users (Daily Commuters and Occasional Riders), City Officials and Policymakers.

1.5.1 IndyGo Transit Planners play a critical role in planning and managing public transit services in Indianapolis. Their primary insight needs include analyzing ridership demand across different timeframes, identifying underperforming routes that may require modifications or increased service frequency, and monitoring peak-hour ridership to adjust fleet allocations. Additionally, they need to compare year-over-year ridership trends and assess how ridership varies across different route categories (e.g., high-frequency vs. low-frequency routes). By leveraging these insights, transit planners can optimize route planning, reduce congestion, and improve service efficiency. For example, a transit planner may use the dashboard to analyze high-ridership routes during peak hours and increase service frequency to prevent overcrowding and delays.

1.5.2 Public Transit Users (Daily Commuters and Occasional Riders) rely on real-time and historical ridership data to make informed travel decisions. Their key insight needs include understanding ridership trends to plan their journeys efficiently, identifying high-traffic stops to optimize transfer points, and comparing real-time vs. historical ridership data to choose the best travel times with minimal congestion. Additionally, commuters benefit from tracking weekly and monthly ridership trends to anticipate peak travel days and adjust their schedules accordingly. For instance, a commuter planning a trip could check which routes have the least congestion during rush hour and select a more convenient travel option.

1.5.3 City Officials and Policymakers play a vital role in urban planning, transportation policies, and public funding allocation. Their insight needs revolve around assessing public transit usage trends to support infrastructure planning, identifying low-ridership areas that may require service expansion, and evaluating ridership fluctuations in response to policy changes such as fare adjustments and new service routes. Additionally, city officials analyze year-over-year ridership data to justify budget allocations for transit improvements and assess the impact of major transit projects, such as dedicated bus lanes and fleet electrification initiatives. For example, a city official analyzing annual ridership trends may advocate for increased public transportation funding to improve transit equity and accessibility in underserved communities. The project’s visualizations comprehensively address these diverse needs, enhancing decision-making for all stakeholders.

2. DATA ACQUISITION

The primary objective of data acquisition is to gather comprehensive transit data that enables detailed insights into passenger activity, route performance, and service efficiency. The dataset for this project is sourced directly from IndyGo’s transit system, provided via Google Drive [6], ensuring its accuracy and relevance for ridership analysis. It includes GPS tracker data, APC (Automatic Passenger Counter) records, and schedule adherence logs, capturing detailed ridership activity from 2022 to 2024 enabling long-term and year-over-year trend analysis. Data is structured into fact and dimension tables and provided in CSV format, ensuring compatibility with tools like Power BI. A comprehensive data dictionary outlines field definitions, data types, and table relationships to support seamless integration. We performed exploratory data analysis to assess total records, detect missing values, and compute statistical summaries, laying the foundation for visualization and interpretation.

2.1 Data Sources (Click Here To Access Data): The dataset is collected from multiple primary sources within IndyGo’s transit system [6], providing accurate and real-time transit data. These sources include:

GPS Trackers: These devices capture real-time bus locations, arrival and departure times, and schedule adherence metrics. The data helps assess on-time performance, identify delays, and track route efficiency over time. **Automatic Passenger Counters (APCs):** These counters record boarding and alighting counts at each stop, providing ridership insights by time of day, route, and service type. This data is crucial for understanding demand patterns and optimizing bus schedules. **Schedule Data:** This dataset contains planned service schedules for each route, allowing for comparisons between scheduled vs. actual performance.

To ensure data quality and consistency, preprocessing steps were applied, including handling missing values, filtering unnecessary columns, and standardizing formats. These steps help maintain the integrity and

usability of the dataset, allowing for seamless integration into Power BI dashboards for visualization.

2.2 Data Description, Quality and Coverage: The dataset follows a relational structure, consisting of 1 fact and 6 dimension tables, which allow for efficient querying and aggregation of data. These were joined using shared foreign key relationships, resulting in a unified dataset suitable for answering our research questions and supporting user insight needs.

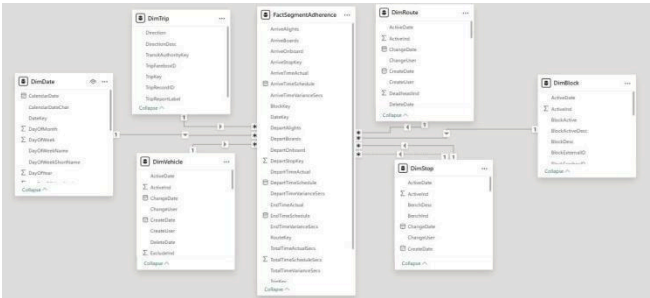
Entities and Major Attributes

Fact Tables: These tables contain quantitative data such as ridership counts, boarding/alighting details, on-time performance metrics, and trip adherence data. The key fact table, FactSegmentAdherence (FSA), includes critical numerical attributes such as arrival times, passenger load, and trip durations, providing a detailed view of service performance.

Dimension Tables: These tables store descriptive attributes, including route details, stop locations, vehicle information, and service categories. Some key dimension tables include:

- DimDate: Provides date-related attributes for trend analysis (e.g., trip date, weekday type).
- DimRoute: Contains route metadata, including route ID, report labels, descriptions and route activation status.
- DimStop: Lists bus stop details, such as stop IDs, descriptions, and coordinates.
- DimTrip: Stores trip-level information, such as trip ID, direction, and farebox ID.
- DimVehicle: Provides vehicle-related attributes, including vehicle ID and labels.
- DimBlock: Contains block-level metadata including block identifiers and labels, useful for analyzing route groupings and expired route information.

Fig. 2 Shows Data Model Diagram - Star Schema (Click Here)



The dataset maintains stop-level granularity, allowing for highly detailed ridership insights. Covering a three-year period (2022–2024), the dataset enables comparative analysis across different time frames, including daily, weekly, and monthly ridership trends. This extensive coverage facilitates the identification of long-term patterns, seasonal variations, and emerging transit trends.

Descriptive Statistics: After merging all the dataset source files, the resulting dataset contained approximately 20.62 million records, each representing a specific trip segment involving details of arrival/departure times, passenger counts, trip duration, and adherence to scheduled service. The merged dataset included around 197 attributes derived from the original 7 tables.

To ensure data quality and completeness, validation checks were performed, including: Handling missing values to ensure consistency in records, standardizing date formats for temporal analysis, merging key attributes across multiple tables to ensure data integrity and removing duplicate records to prevent inconsistencies in analysis. By implementing these pre-processing steps, the dataset is now well-structured, high-quality, and ready for use in IndyGo’s Power BI Ridership Dashboard. Overall, the dataset’s quality and extensive coverage support the project’s analytical goals, providing reliable insights for stakeholders.

Table 1 – Summary of Dataset Source

Dataset Source File Name	Rows	Columns
FactSegmentAdherence.csv	20,618,440	27
DimBlock.csv	880	20
DimDate.csv	1,096	19
DimRoute.csv	1,236	44
DimStop.csv	19,349	42
DimVehicle.csv	2,570	38
DimTrip.csv	226,132	7

3. DATA ANALYSIS

The analysis of IndyGo ridership data involves multiple statistical and visualization techniques to uncover insights related to ridership trends, route performance, and stop-level activity. The data analysis process started with the creation of an Entity-Relation (ER) diagram to better understand the relationship between different tables. The objective is to transform raw data into actionable insights that address key research questions and client needs. The initial analysis was conducted using Python for data processing and Power BI for interactive visualizations, ensuring effective data exploration and representation.

3.1 Analytical Techniques: To extract meaningful insights, a combination of exploratory data analysis (EDA), statistical analysis, and data visualization techniques were employed. These methods were selected to align with the project's objectives and provide a comprehensive understanding of ridership patterns.

Exploratory Data Analysis (EDA): Data Cleaning and Preprocessing: The dataset underwent handling of missing values, removal of duplicates, and standardization of date formats to ensure consistency and reliability. **Feature Engineering:** New attributes such as daily ridership trends, route-based categories, and peak-hour indicators were created to enhance the analysis. **Descriptive Statistics:** Mean, median, standard deviation, and quartile analyses were performed on ridership metrics to understand data distributions and variations.

3.2 Tools and Libraries: The data analysis and visualization process utilized a combination of Python and Power BI to support efficient data handling and insightful reporting. Python was primarily used for data preprocessing, statistical analysis, and initial visual exploration, while Power BI facilitated the creation of interactive dashboards. Within Python, Pandas enabled essential data manipulation tasks, including merging, filtering, and aggregation. NumPy was applied for statistical computations to derive descriptive metrics. For early visualization of trends, Matplotlib and Seaborn were employed to produce line charts, bar graphs, scatter plots, and heatmaps illustrating ridership variations over time. After preprocessing, Power BI was used extensively to build a robust, interactive dashboard. It supported real-time filtering, drill-down analysis, and dynamic reporting. A variety of visual elements—line charts, bar charts, heatmaps, pie charts, and KPI cards—were created to communicate insights related to ridership patterns, route performance, and stop-level demand. The integration of these tools ensured a seamless analytical workflow, transforming raw data into actionable insights.

4. VISUALIZATIONS

The [IndyGo Ridership Power BI Dashboard](#) utilizes a range of visualizations to convey ridership patterns clearly and effectively. These visuals are aligned with the project's research questions and tailored to support stakeholder decision-making. Each was selected to highlight key trends across timeframes, routes, and stops, offering actionable insights and a comprehensive view of transit usage.

4.1 Ridership Trends over Time: To analyze ridership patterns across different timeframes, multiple visualizations are employed. A line chart depicting daily ridership trends illustrates fluctuations in ridership per day over a month, enabling stakeholders to observe peaks and dips in commuter volume. A bar chart displaying weekly ridership trends highlights variations in passenger count by day of the week, helping transit planners optimize service schedules. Additionally, a monthly ridership trends bar chart offers insights into seasonal ridership changes, while a yearly ridership comparison evaluates total ridership trends over 2022-2024 through multi-card. The dashboard also includes filtering options for months, years, and route descriptions, allowing for deeper

insights into specific transit patterns. These visualization methods help in identifying trends, seasonal variations, and daily commuting behaviors.

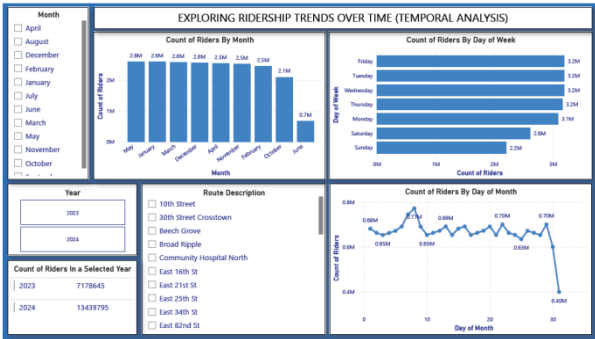


Fig. 3 Shows Ridership Trends over Time - [Click Here](#)

4.2 Changes in Ridership Trends across Different Time Frames: This visualization set focuses on short-term and medium-term trends in ridership. A bar chart visualizing change in ridership between 2023 and 2024 helps assess growth or decline trends across day of the week, showing which days experienced significant shifts in passenger volume. A weekly ridership trends line chart showcases fluctuations in total ridership over the last five weeks, highlighting short-term changes in transit use. Additionally, a heatmap depicting daily ridership patterns for different weekdays over the last five weeks provides granular insights into week-to-week variations using color intensity to highlight high and low ridership periods, allowing planners to make short-term service adjustments. Moreover, a route selection panel allows filtering of ridership trends by specific transit routes, enabling a more granular analysis.

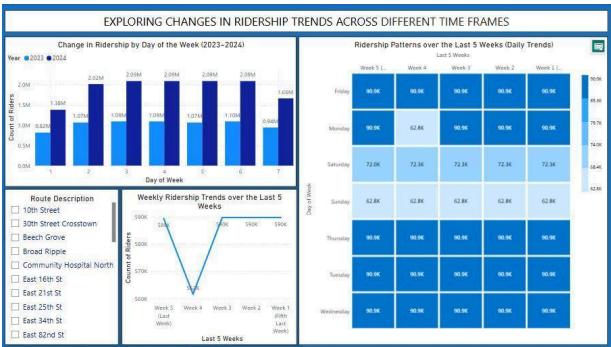


Fig. 4 Shows Changes in Ridership Trends across Different Time Frames - [Click Here](#)

4.3 Ridership Variation across Different Route Categories: Understanding how ridership varies based on service frequency is crucial for optimizing transit operations. Donut chart depicts total ridership distribution across different routes, highlighting key routes used by commuters. A pie chart representing total ridership and percentage share by route category (low, medium, and high bus service frequency) allows for quick comparisons of ridership distribution across different service levels which help planners to assess which routes require additional resources. Lastly, a heatmap visualizing ridership volume across route categories and time of day enables stakeholders to optimize service during peak hours.

4.4 Ridership Trends for Specific Routes over the Past 12 Months: To monitor long-term trends in ridership for individual routes, a line chart illustrates ridership trends over the last 12 months for selected routes. This helps stakeholders to identify seasonal fluctuations, peak travel months, and potential service inefficiencies. By tracking year-over-year performance, transit planners can adjust scheduling, introduce new routes, modify underutilized routes to improve efficiency.

4.5 Stops with the Highest and Lowest Ridership: KPI visualization showcases the highest and lowest ridership bus stops, providing a concise summary of top-performing and underutilized bus

stops. This insight assists in optimizing stop locations, improving service coverage, and identifying transit deserts that may require intervention.

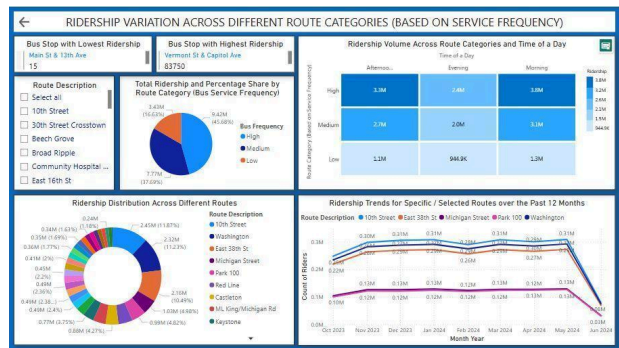


Fig. 5 Shows Ridership Variation across Different Route Categories, Ridership Trends for Specific Routes over the Past 12 Months, Stops with the Highest and Lowest Ridership - [Click Here](#)

5. INTERPRETATION OF RESULTS

The analysis of IndyGo’s ridership data reveals several important trends. Weekdays (Tuesday to Friday) consistently show the highest commuter volume, averaging ~3.2 million riders per day, while weekends, particularly Sundays, experience the lowest (~2.2 million). Monthly trends indicate that January, March, and December have the highest ridership (~2.6M/month), whereas June shows a significant drop to just 0.7 million, suggesting possible seasonal or operational effects. On a daily scale, ridership peaks at 0.77 million, with lows reaching 0.40 million, especially toward the end of the month. A year-over-year comparison shows substantial growth, with total ridership increasing from 7.1 million in 2023 to 13.4 million in 2024. This reflects a rising reliance on public transportation and emphasizes the need for data-driven planning. Ridership in 2024 showed consistent growth across all weekdays and weekends compared to 2023. The most stable high-volume days remained Tuesday through Thursday, while Saturdays and Sundays continued to have lower ridership. However, weekend usage surged significantly, with Sunday rising from 0.94M to 1.66M and Saturday from 0.82M to 1.38M. Weekdays also experienced average gains of ~0.95M to 1M riders per day, with Tuesday peaking at 2.09M. A temporary drop in the second week of May 2024, followed by a quick rebound, suggests potential external disruptions. Heatmap visualizations further confirmed that weekday ridership holds steady around 2.7M, while weekend usage remains lower. In terms of operations, high-frequency routes accounted for 45.68% (9.42M) of total ridership, with Vermont St & Capitol Ave seeing the highest usage (83,750 riders) and Main St & 13th Ave the lowest (15 riders). Ridership peaks during morning and afternoon hours, emphasizing its importance for daily commutes. 10th Street, Washington, and East 38th St routes made up over 34% of ridership, consistently drawing ~0.29M–0.31M monthly riders from Oct 2023 to May 2024. In contrast, Michigan Street and Park 100 maintained lower but steady figures (~0.12M–0.13M). A system-wide decline in June 2024 may indicate seasonal shifts or operational changes that merit further exploration. These findings provide valuable direction for IndyGo to optimize service schedules, allocate resources more efficiently, and enhance accessibility based on observed commuter behaviors. These findings also support strategic planning to adjust service frequency, optimize stop locations, and enhance overall system efficiency and equity.

6. EVALUATION AND REDESIGN

To validate the effectiveness of our visualizations and gain insights into their clarity and usability, we conducted informal evaluations with peers, colleagues, and stakeholders. Feedback was gathered on the clarity of visualization elements, accessibility of information, and overall ease of interpreting trends.

6.1 Evaluation Feedback and Redesign: Most users appreciated the structure and layout of the dashboard, especially the weekly ridership line chart, the daily ridership heatmap, and the percentage change bar chart across 2023–2024. These visualizations effectively communicated temporal ridership trends and highlighted short- and medium-term fluctuations. However, some evaluators shared the following concerns:

Ambiguity in Labels and Legends: A few users found some axis labels and legends unclear, especially in the heatmap. **Redesign Solution:** We improved all axis titles and added tooltips to ensure better readability. **Legends were explicitly titled to reduce interpretation errors.** **Color Overload in Heatmaps:** Some viewers felt overwhelmed by the similar color intensities across days and weeks. **Redesign Solution:** We optimized the color gradient to enhance contrast between high and low ridership values, allowing for clearer differentiation of trends. **Route Filtering Overlap:** Some users were unclear whether filters were applied or not when interacting with route selection. **Redesign Solution:** We introduced a dynamic title/sub-header that updates to reflect the active route being analyzed.

6.2 Challenges Encountered: Data Quality and Consistency: Several datasets contained missing values and inconsistencies across stop keys and vehicle IDs. We resolved this through thorough preprocessing, including deduplication, standardization, and merging across fact-dimension tables using validated keys. **Complex Join Operations:** Joining the fact table with six dimension tables using inner joins required careful handling of null and unmatched records to avoid data loss. Optimized join logic ensured robust merging while preserving essential fields for visualization. **Scalability of Visuals:** Some visuals became cluttered when displaying all routes or full-month data at once. We introduced interactive filtering and dynamic visualization titles to address scaling concerns and support a more tailored view. **Opportunities for Enhancement:** **Temporal Event Overlays:** In future iterations, integrating external event data (e.g., weather, city events, or disruptions) could help explain anomalies like ridership dips or surges. **User-Specific Dashboards:** Role-based dashboards for planners, policymakers, and commuters could improve personalized insights and decision-making. **Predictive Analytics Integration:** A natural extension of our work could involve forecasting models (e.g., time-series forecasting with ARIMA or Prophet) to anticipate ridership trends. **Mobile-Friendly Design:** As a long-term goal, transforming the dashboard into a mobile-responsive version would increase accessibility for real-time users such as daily commuters.

USAGE AND CRITIQUE OF AI TOOLS

AI tools were used in this project to support report refining and visual communication. A visual abstract was generated using AI to summarize the project’s goals, data, tools, and key insights. Language models also assisted in refining content throughout the report. While these tools improved efficiency and clarity, they have limitations, including potential biases, lack of domain-specific context, and limited design flexibility. Human oversight was essential to validate and tailor outputs. For future projects, AI can enhance productivity, but should be used critically alongside expert input.

ACKNOWLEDGEMENTS

The author expresses gratitude to Project Sponsor Satya Sarfare, Professor Katy Börner, and Professor M. Ginda, for their invaluable guidance throughout the project course. Appreciation is extended to all Associate Instructors for their expert insights and constructive feedback during the project development. Additionally, the author acknowledges IndyGo for providing their transit system data.

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- Dataset: IndyGo’s Transit System, [IndyGo IU InfoViz Project Data - Google Drive](#)
- IndyGo Ridership Power BI Dashboard - [Final Project Results](#)

